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📘 MCP (Model Context Protocol) Quiz

Total Questions: 20
Type: MCQ + Scenario-Based + Short Answer
Difficulty: Medium to Advanced

🧠 Section A: Multiple Choice Questions (MCQs)

Q1. What is the primary purpose of the Model Context Protocol (MCP)?
A. Data storage
B. Standardized agent-tool communication
C. Model training
D. UI rendering

Q2. Which MCP integration pattern focuses on delegating tasks across multiple agents?
A. Single-agent pattern
B. Pipeline pattern
C. Multi-agent orchestration
D. Batch processing

Q3. In MCP, tool invocation typically follows which structure?
A. Direct function call
B. API request with structured schema
C. Manual input
D. File upload

Q4. Which component ensures that agents can safely share data?
A. Tokenization

B. Secure context sharing
C. Logging
D. Serialization

Q5. What is a key advantage of MCP in enterprise environments?
A. Faster UI rendering
B. Vendor lock-in
C. Interoperability across tools
D. Reduced storage

Q6. Which integration pattern is best suited for sequential task execution?
A. Event-driven pattern
B. Pipeline pattern
C. Mesh pattern
D. Parallel pattern

Q7. What does “context” refer to in MCP?
A. UI design
B. Model weights
C. Data passed between agents/tools
D. Network latency

Q8. Which of the following improves security in MCP?
A. Hardcoding credentials
B. Context encryption
C. Open access APIs
D. Public tokens

⚙️ Section B: Scenario-Based Questions

Q9. A financial company uses multiple AI agents for fraud detection, reporting, and alerting. Which MCP pattern is most suitable?
A. Single-agent
B. Multi-agent orchestration

C. Static pipeline
D. Batch processing

Q10. An AI system fetches stock data, analyzes it, and generates a report in sequence.
Which pattern is being used?
A. Parallel processing
B. Pipeline pattern
C. Mesh network
D. Event loop

Q11. An agent calls an external weather API using MCP.
What is this process called?
A. Context sharing
B. Tool invocation
C. Model tuning
D. Serialization

Q12. A healthcare system ensures patient data is encrypted before sharing between agents.
Which concept is applied?
A. Data batching
B. Secure context sharing
C. Logging
D. Caching

Q13. A chatbot uses MCP to connect with CRM, database, and analytics tools.
What is this an example of?
A. Standalone AI
B. Enterprise agent integration
C. Local processing
D. Static system

✍️ Section C: Short Answer Questions

Q14. Explain the difference between *pipeline pattern* and *multi-agent orchestration* in MCP.

In Pipeline the tasks done in sequence (step-by-step).
But Multi-agent orchestration the multiple agents collaborate dynamically.

Q15. Describe how tool invocation works in MCP architecture.

Agent sends structured request → MCP routes it → tool/API executes → response returned.

Q16. What are the risks of insecure context sharing? Mention at least two.

- [ ] Data leakage
- [ ] Unauthorized access
- [ ] Privacy violations

Q17. How does MCP improve scalability in enterprise AI systems?

MCP enables modular agents + tools → easy scaling by adding more agents/services.

🧩 Section D: Case Study / Analytical

Q18.
A retail company wants to build an AI system that:

- Retrieves customer data
- Recommends products
- Sends personalized emails

Question:
Design an MCP-based integration approach and justify your chosen pattern.

Approach: Use **Pipeline Pattern with MCP**

Flow:

1. Agent 1 → retrieves customer data (CRM/database)
2. Agent 2 → generates product recommendations (ML model)
3. Agent 3 → sends personalized emails (email service)

We use Pipeline because:

- Tasks are **sequential and dependent**
- Output of one step → input to next
- Easy to manage and scale

Q19.
An organization is concerned about data privacy when multiple agents interact.

Question:
Suggest strategies using MCP to ensure secure context sharing.

Strategies using MCP:

Encryption → protect data during transfer

Role-Based Access Control (RBAC) → limit who can access what

Authentication & Authorization → verify agents/tools

Audit Logs → track data usage

Data minimization → share only required data

Result: Prevents leaks, ensures privacy, and maintains compliance

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Q20.
Compare MCP-based architecture with traditional API-based integration in enterprise systems.

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- **Encryption** → protect data during transfer
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Result: Prevents leaks, ensures privacy, and maintains compliance

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